Date 7 November 2026
To [Client name], [Watershed association]
From Team [4]: [Member one], [Member two], [Member three]
Subject **Customer needs and specifications for a low-cost river turbidity sensor**

Introduction

Muddy water is more than a nuisance in a small river. Suspended sediment carries nutrients and pollutants downstream, buries the gravel where fish spawn, and raises the cost of treating drinking water. Most of this sediment moves during a few storms each year. Volunteer groups, however, can afford only weekly grab samples, and these seldom catch a storm.

The [Watershed association] has asked our team to design a turbidity sensor that its volunteers can build, install and maintain for less than $150 a site. In this report, a *customer* is a volunteer who installs and services the sensor. The association's staff, who analyse the data, are secondary customers.

The next section presents the customer needs that we found. The section after it turns each primary need into a metric with a target value and a test. Appendix A gives our schedule for the rest of the project.

Customer Needs

We identified the needs through interviews with four volunteers and two staff members, visits to two monitoring sites, and the project description [1]. We then grouped them into primary needs, which the design must meet, and secondary needs, which would improve it.

Primary Needs

The sensor must stay accurate during storms. Staff members want readings that they can compare with laboratory measurements made by the ISO 7027 method [2]. Because turbidity can rise from below 20 NTU to several hundred NTU within hours, the sensor must cover that whole range.

The sensor must cost less than $150 per site. The association's grant covers forty sites at that price. At the price of one laboratory instrument, the same grant would cover a single site.

Volunteers must be able to service the sensor in the field. Most volunteers have no training in electronics, and they visit each site about once a month. Cleaning the optical window and replacing the battery must therefore take minutes and need no tools.

Secondary Needs

Volunteers would also like the sensor to send its readings over the phone network, so that they need not download data on site. They would also like it to be inconspicuous, so that it is less likely to be stolen or damaged.

Metrics and Specifications

Table 1 links each primary need to a metric, a target value, the source of that target, and the test that will show whether a prototype meets it.

Table 1. Metrics and target values for the primary needs

Need	Metric	Target	Source of target	Test
Accurate in storms	Difference from a laboratory turbidimeter	≤ 5 % over 0–1000 NTU	Staff interviews; range of ISO 7027 [2]	Formazin standards at eight levels
Low cost	Cost of parts per site	≤ $150	Grant budget for forty sites	Bill of materials
Field service	Time to clean the window and change the battery	≤ 10 min, no tools	Volunteer interviews	Three volunteers, timed

The accuracy target needs the most care. A difference of 5 % is about the spread between two laboratory instruments that measure the same sample. A tighter target would therefore test the reference instrument rather than our sensor.

Conclusion

Volunteers need a sensor that stays accurate through storms, costs less than $150 and can be serviced without tools; staff members need data that they can compare with laboratory measurements. In the next phase, we will generate concepts for the optics and the housing and select one against the targets of Table 1. Appendix A shows when each phase ends.

Appendix A: Project Schedule

Table A-1 lists the phases of the project and their end dates. We chose the lead for each phase by experience: [Member one] has built optical circuits, [Member two] has worked with data loggers, and [Member three] has run field campaigns.

Table A-1. Phases of the project, with their leads and end dates

Phase	Lead	Ends
Customer needs and specifications	[Member three]	Week 4
Concept generation and selection	[Member one]	Week 7
Prototype and laboratory tests	[Member two]	Week 11
Field test and final report	[Member three]	Week 15

References

[1] [Course instructor], “Project description: a low-cost turbidity sensor for volunteer monitoring.” 2026.

[2] ISO, “Water quality – Determination of turbidity – Part 1: Quantitative methods,” Geneva, technical report ISO 7027-1:2016, 2016.